Emotional Adaptive Navigation

Revised Research Protocol & Implementation Plan

Core idea: test distinct menu structures under different emotional contexts, identify which structures work best for each context, and use those findings to drive emotion-contingent navigation.

Decision	Selected direction
Platform	Simulated e-commerce website / fictional watch store
Navigation structures	Horizontal menu, vertical sidebar, mega menu, and fractal/hierarchical menu
Emotion measurement	Interaction signals plus brief self-reports; no webcam or facial analysis
Research scope	Three linked studies: layout-by-emotion mapping, controlled adaptive-menu test, end-to-end system evaluation

Important design principle. The project does not assume that a particular layout is best for a particular emotion. Study 1 estimates the mapping empirically; Study 2 tests whether using that mapping causally helps; Study 3 tests the complete system with imperfect emotion estimates.

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1. Executive summary

This project investigates whether an e-commerce website can improve product-finding by adapting its navigation structure to a user's current emotional context. Rather than only changing link priority or adding help text inside one fixed menu, the central experiment compares several genuinely different menu structures and asks which structure performs best under each measured emotional state.

The test environment will be a fictional online watch store. Participants will complete controlled product-finding tasks using four candidate structures: a horizontal navigation bar, a vertical sidebar, a mega menu, and a fractal/hierarchical menu inspired by adaptive hierarchical navigation. The four emotional contexts are positive affect, frustration, confusion/lostness, and boredom/disengagement.

The research is organised into three studies. Study 1 establishes a performance matrix across menu structures and emotional contexts. Study 2 tests whether adapting to the Study 1 mapping outperforms a static menu and an emotion-mismatched menu. Study 3 evaluates a live system that estimates state using interaction signals anchored by brief self-reports and applies the selected structure at safe task boundaries.

Sample sizes are provisional planning targets. A pilot and a formal power analysis are required before preregistration. The full factorial design may need to be simplified if pilot timing, fatigue, or power analysis shows it is too demanding.

What makes this project distinctive

- **Structure is the main adaptation.** The system can switch among different navigation architectures, not merely reorder links within a fixed architecture.
- **Emotion-to-layout mapping is discovered experimentally.** No layout is pre-labelled as the correct choice for frustration, confusion, boredom, or positive affect.
- **Emotion inference is treated cautiously.** Clicks and pauses are behavioural proxies, not objective proof of emotion; brief self-reports provide the primary state labels.
- **Adaptation safety is part of the design.** Whole-menu changes happen between tasks or at an explicit transition, not abruptly while a participant is navigating toward a target.

2. Research questions and hypotheses

Primary research question

Which e-commerce navigation structure best supports product-finding in each of four emotional contexts, and does adapting the structure according to that empirically derived mapping improve performance and user experience compared with static or mismatched navigation?

Secondary research questions

- Do the four menu structures differ in completion time, task success, navigation errors, backtracking, and perceived usability?
- Do the relative advantages of the menu structures change across emotional contexts, or is one structure consistently best regardless of state?
- Can interaction features predict self-reported emotional context on held-out participants better than a majority-class baseline?
- Does emotion-contingent structural adaptation remain beneficial when the system's emotion estimate is imperfect?
- Does changing structures create disorientation, loss of spatial memory, or lower user control?

Hypotheses

ID	Hypothesis	Primary evidence
H1	Menu structure and emotional context will interact: relative layout performance will differ across at least some contexts.	Structure × context interaction in mixed-effects model
H2	A policy that selects the layout supported by Study 1 will improve task performance versus a fixed layout.	Matched adaptive vs static contrast in Study 2
H3	Emotion-matched adaptation will outperform emotion-mismatched adaptation.	Matched vs mismatched contrast
H4	Interaction signals will predict self-reported state above a majority-class baseline, but with meaningful uncertainty.	Participant-grouped validation; macro-F1 and confusion matrix
H5	End-to-end benefits will be smaller than the oracle/known-label test and will depend on estimator confidence and transition stability.	Study 3 compared with Study 2

3. Emotional taxonomy and measurement

Use four operational emotional contexts. They are simplified experimental categories, not a claim that real emotional experience is mutually exclusive or can be read directly from clicks. Record self-report ratings as continuous values where possible, then use a preregistered rule to identify the dominant context for condition assignment and analysis.

State	Operational definition	Potential behavioural signals (not proof)
Positive affect	Participant reports positive valence / good mood after a brief positive-feedback or successful-task induction.	More direct choices, fewer reversals; may also reflect familiarity or confidence.
Frustration	Participant reports irritation or frustration following a mild, controlled obstacle or unsuccessful interaction.	Repeated clicks, rapid reversals, repeated menu open/close, pauses after failure.
Confusion / lostness	Participant reports uncertainty about where a product belongs or which navigation path to choose.	Longer pauses, category switching, backtracking, repeated menu inspection, inconsistent paths.
Boredom / disengagement	Participant reports low interest or under-stimulation after a repetitive, low-challenge pre-task activity.	Long idle gaps, unusually fast guesses, reduced exploration, lower engagement ratings.

Measurement protocol

- Collect brief state ratings immediately after each emotion induction and at selected trial boundaries; keep the questionnaire short to reduce interruption.
- Use a consistent rating scale for valence and discrete items for frustration, confusion, and boredom. Pilot the wording and check whether participants understand the distinctions.
- Log timestamps, menu openings, hover/focus events where available, clicks, misclicks, backtracks, category switches, search use, task success, and completion time.
- Do not label a single pause or backtrack as a particular emotion. Use multiple features, self-report anchors, and uncertainty estimates.
- No webcam, facial analysis, or biometric sensor is required for the core plan. Eye tracking may be considered only as a separately scoped extension if equipment and approvals are available.

Emotion induction and ethics

Use mild, reversible inductions embedded in the task sequence. Avoid humiliating, highly stressful, or personally sensitive material. Participants may stop at any time. Debrief them that emotional states were studied as experimental contexts and that interaction signals are imperfect indicators. The protocol should be reviewed under the institution’s applicable ethics process before recruitment.

4. Experimental platform: fictional watch store

Build a controlled web prototype with approximately 100–120 fictional watches and a consistent product taxonomy. Product attributes can include price, intended use, style, strap material, case size, movement type, and water resistance. Avoid real logos and recognisable brands so prior brand familiarity does not dominate the task.

Information architecture

- Define the same categories, labels, products, and destination pages for every menu condition.
- Use a stable hierarchy of top-level categories, subcategories, and product-level filters; ensure each target product has an unambiguous route.
- Create matched tasks across difficulty levels, such as finding a dress watch below a budget, a sports watch with a required feature, or a product matching a material/size constraint.

- Keep visual styling, typography, colour, product cards, content, and page loading behaviour as similar as possible across structures.
- Log task ID, target ID, assigned menu structure, emotional context, condition order, and every navigation event.

Candidate menu structures

Structure	Implementation definition	Design controls
A. Horizontal bar	Top-level categories appear in a horizontal top navigation; subcategories open through a consistent dropdown or second level.	Same labels and hierarchy; no extra categories.
B. Vertical sidebar	Persistent vertical category tree on the left, with expandable/collapsible branches.	Control sidebar width, nesting indicators, and expansion behaviour.
C. Mega menu	A top-level category opens a larger panel showing several related subcategories and selected links at once.	Keep visible link inventory and grouping comparable; avoid adding information absent elsewhere.
D. Fractal / hierarchical menu	Progressive, focused navigation through nested levels, with clear parent/child relations and a consistent way to move back or zoom out.	Specify exact interaction before implementation; preserve predictable backtracking and labels.

The fractal menu must be operationally defined before data collection. The name alone is not enough: document its visual states, navigation transitions, hierarchy depth, and recovery path so it can be reproduced and fairly compared.

5. Overall study design

The studies answer different questions and should not be collapsed into a single test. Study 1 discovers the layout-by-emotion performance pattern; Study 2 tests the causal value of using that pattern; Study 3 evaluates a realistic estimator and transition policy.

Study	Main question	Design / planning target	Output
Pilot	Are tasks, inductions, and layouts understandable and tolerable?	6–8 participants; small usability pilot.	Refined protocol, timing, task difficulty, technical fixes.
Study 1: mapping	Which structure works best in each emotional context?	Within-participant layout × emotion design; planning target around 30 participants, subject to power analysis and fatigue.	Performance matrix and candidate emotion-to-layout policy.
Study 2: causal test	Does applying the mapping improve outcomes?	Within-participant comparison of static, emotion-matched, and emotion-mismatched policies; planning target 36–48.	Causal estimate of structural adaptation benefit.
Study 3: end-to-end	Does the system still help when state is estimated imperfectly?	Feasibility study of static vs live hybrid adaptation; planning target 24–36.	Estimator performance, real-world benefit, transition/disorientation evidence.

Design burden and feasibility

Four structures × four emotional contexts creates 16 structure-context cells before adding repeated tasks, order counterbalancing, or comparison conditions. A fully repeated-measures Study 1 may be tiring and vulnerable to learning effects. The pilot should establish session length. If needed, use a balanced incomplete-block design or split the design across sessions while ensuring each cell has sufficient observations.

Counterbalancing and controls

- Counterbalance layout order and task order using a Latin-square or balanced sequence where feasible.
- Use multiple matched tasks per cell so one unusually easy or hard target does not determine a layout’s ranking.
- Record previous exposure to each structure, task difficulty, trial number, and order for analysis.
- Use identical products, labels, hierarchy, and task instructions across layouts.
- Predefine how ties will be resolved when two structures perform similarly; do not select the winner based only on the smallest mean time.

6. Pilot study

Recruit 6–8 participants to test the prototype and protocol, not to make confirmatory claims. Check that all four layouts are usable, emotional inductions produce the intended self-reported variation without excessive distress, tasks have an appropriate difficulty range, event logs are complete, and the full session length is acceptable.

Pilot checklist

- Verify every target can be reached in each layout and all menu controls work with mouse and keyboard.
- Check task wording does not accidentally reveal the route or category.
- Measure induction effectiveness using self-report; revise weak or confusing inductions.
- Inspect whether the four structures are sufficiently distinct yet comparable in information content.
- Estimate within-person variability and missingness to inform power analysis and sample size.
- Use participant feedback to identify visual or interaction confounds before the main studies.

7. Study 1 — map emotional context to menu structure

Objective: empirically determine how the four menu structures perform under each of the four emotional contexts. This is the key study that directly operationalises the original idea.

Design

Use a repeated-measures design where each participant experiences multiple menu structures and emotional contexts. Ideally, each participant contributes observations to all 16 cells, but use a balanced incomplete-block or multi-session design if fatigue or learning would undermine data quality. The emotional context is induced or prompted before the relevant task, then checked with a brief self-report. The order of structures, states, and tasks must be counterbalanced.

Procedure per trial

- Show a short instruction and apply the planned mild emotional-context induction.
- Collect a brief state rating to check whether the intended context was experienced.
- Present the assigned menu structure and a product-finding task using a matched target.
- Record completion time, success, errors, backtracking, menu interactions, and any search use.
- Collect a short usability/difficulty rating and, where scheduled, a brief state re-check.
- Provide a neutral reset or break between inductions to reduce carry-over.

Primary outcomes

Outcome	Operationalisation	Role
Task success	Correct target reached within a defined time limit.	Primary / safety check
Completion time	Time from task start to correct target; unsuccessful trials handled by preregistered rule.	Primary efficiency measure
Navigation errors	Misclicks, wrong branches, or invalid route transitions.	Secondary
Backtracking	Returning to a previously visited category/level or reversing route.	Secondary
Perceived usability / workload	Short post-task rating(s) with consistent wording.	Secondary
Self-reported emotional context	Valence and discrete frustration/confusion/boredom ratings.	Manipulation check / state label

How to identify the best structure

Analyse the layout × emotional-context interaction. Estimate the marginal performance of each structure within each context, with participant and task variability accounted for. The selected mapping should consider success first, then efficiency and errors, then subjective usability. If a layout is fastest but meaningfully increases failures or disorientation, do not label it the winner. Where differences are small or uncertain, treat the mapping as inconclusive or allow more than one acceptable layout.

Study 1 deliverable

A documented 4 × 4 performance matrix, confidence intervals, and a preregistered selection rule producing the candidate mapping used in Study 2. The mapping is an empirical result, not a prewritten assumption such as “frustrated users need a sidebar.”

8. Study 2 — test emotion-contingent structural adaptation

Objective: test whether using the Study 1 mapping improves navigation, rather than merely confirming that layouts differ.

Conditions

Condition	What the participant experiences	Purpose
1. Static structure	One preselected menu structure remains in use across emotional contexts. Select the baseline before confirmatory testing (e.g., the best overall structure from Study 1, not a post-hoc convenient choice).	Baseline for ordinary non-adaptive navigation.
2. Emotion-matched adaptation	At safe task boundaries, the system selects the structure supported by Study 1 for the current self-reported/known experimental context.	Test whether the mapping is useful.
3. Emotion-mismatched adaptation	The system selects a structure mapped to a different context according to a preregistered mismatch rule.	Test whether matching context matters rather than any change helping.

Use known/checked emotional labels in this study where possible. That isolates the effect of structural adaptation from errors made by an emotion estimator. The mismatch rule must be fixed before analysing outcomes; avoid choosing the most harmful mismatch after seeing the data.

Procedure and controls

- Use new but difficulty-matched tasks to reduce memorisation from Study 1.
- Counterbalance the three conditions and task order.
- Keep all menu content and product information constant; the intended independent variable is the structural policy.
- Make transitions visible and predictable. Apply a new structure at the start of the next task, not mid-route.
- Measure task performance, usability, perceived predictability, disruption, and whether the participant noticed or understood the change.

Primary contrasts

- Emotion-matched adaptation vs static structure: does adaptation improve performance without reducing success?
- Emotion-matched vs emotion-mismatched adaptation: does correct matching matter?
- Static vs mismatched adaptation: is arbitrary structural switching harmful or merely neutral?

Interpretation

If matched adaptation does not outperform the static baseline, the correct conclusion may be that structural switching offers little benefit, that one structure works well across states, or that switching costs offset any state-specific benefit. Such outcomes are valuable and should not be hidden. Study 2 is the primary causal test of the original idea.

9. Study 3 — end-to-end adaptive navigation

Objective: evaluate the complete system using interaction signals plus brief self-reports to estimate emotional context, select a structure using the Study 1 mapping, and apply it without unnecessary disruption.

System pipeline

Step	System behaviour
1. Observe	Collect low-cost interaction features: pauses, backtracks, menu open/close, misclicks, category switching, and task progress.
2. Ask / anchor	At predefined boundaries, collect a brief self-report; the design should not repeatedly interrupt users after every click.
3. Estimate	Combine interaction features and self-report where available. Return a state distribution or confidence score rather than only a hard label.
4. Select	Use the frozen Study 1 mapping to choose a candidate structure for the estimated context.
5. Transition safely	Offer the structural change between tasks or at a clear checkpoint; explain the change, allow the user to keep the current layout or revert.
6. Evaluate	Log estimate, confidence, selected structure, user acceptance/reversion, task outcomes, and any reported disorientation.

Conditions

- **Static baseline:** the same chosen structure remains throughout the session.
- **Hybrid adaptive:** the system uses the interaction-signal + self-report estimator and the frozen emotion-to-structure mapping.
- If sample size and time permit, add an optional comparison in which only functional presentation changes within the current structure (e.g., clearer guidance or prioritized links). Treat this as a secondary extension, not a replacement for structural adaptation.

Adaptation safety and user control

- Do not replace the entire menu while the participant is part-way through a navigation path.
- Switch only at a task boundary or explicit transition, preserve the current target and progress, and provide a visible explanation.
- Provide a clear “keep current layout” or revert option; record its use rather than treating it as participant error.
- Avoid rapid oscillation: use a confidence threshold, minimum dwell period, and/or require repeated evidence before switching.
- If confidence is low, keep the current structure and ask a brief self-report at the next appropriate boundary.

Study 3 outcomes

- Estimator: macro-F1, balanced accuracy, confusion matrix, calibration/confidence and participant-grouped validation.
- Navigation: task success, completion time, errors, backtracking and search use.
- Experience: perceived control, predictability, usability, workload, disorientation and preference.
- Adaptation behaviour: number of switches, accepted/reverted changes, time from estimate to transition, and low-confidence fallback frequency.

A hybrid estimator should be evaluated on held-out participants, not random trials from the same participants split across train and test. Otherwise, the model may learn person-specific habits and appear more generalisable than it is.

10. Implementation plan

Recommended implementation

Build the prototype as a web application with a shared product catalogue and a navigation component that can render each of the four structures. Use a common event schema so that interaction logs are comparable across layouts. Keep the emotion-induction logic separate from the menu rendering and the adaptation policy.

Module	Responsibilities
Product catalogue	Product IDs, attributes, categories, filters, target products, and task definitions.
Layout renderer	Render horizontal, sidebar, mega-menu, and fractal/hierarchical structures from the same taxonomy.
Task runner	Present task instructions, start/stop timers, record success/failure, and manage breaks/reset.
Emotion check-in	Collect brief state ratings and associate them with task/trial IDs.
Event logger	Timestamped events using one consistent schema; include participant pseudonym, condition, task, layout, state prompt, and event details.
Adaptation policy	In Study 2, use the frozen mapping and known labels; in Study 3, use estimator output, confidence threshold, safe transitions and revert handling.
Analysis pipeline	Clean logs, derive outcomes, run preregistered models, produce layout-by-state matrix and confidence intervals.

Example event schema

Each event should contain a timestamp, anonymous participant ID, study, trial ID, task ID, emotional-context prompt/label (where applicable), active layout, event type, element/category ID, and optional event-specific value. Avoid collecting names, emails, or unrelated personal information in the research log.

Build order

- 1. Implement the product taxonomy and target tasks before styling menus.
- 2. Implement the four menu structures over the same data and verify functional parity.
- 3. Add event logging and automated checks for missing or impossible routes.
- 4. Add emotion check-ins and induction/task sequencing.
- 5. Implement Study 1 randomisation and exportable data.
- 6. After Study 1 analysis, freeze the mapping and build Study 2 policy conditions.
- 7. Add estimator and safe transition behaviour for Study 3 only after the first two studies are stable.

11. Analysis plan

Predefine outcome handling, exclusions, and the structure-selection rule before confirmatory analysis. Analyse task success and efficiency jointly enough to avoid selecting a layout based on speed alone.

Study 1

- Model completion time using a suitable mixed-effects model (often log-transformed if strongly right-skewed), with menu structure, emotional context, and their interaction as fixed effects.
- Include participant and task/target as random effects where supported by the data; include order/trial number as covariates if prespecified.
- Analyse success using a mixed-effects logistic model; analyse counts such as errors/backtracks with an appropriate count model or robust alternative.

- Report effect estimates and confidence intervals, not only p-values. Correct for multiple comparisons in follow-up contrasts.
- Generate estimated marginal means for every structure × context cell and apply the preregistered selection rule.

Study 2

- Use mixed-effects models with condition as the main fixed effect and participant/task as grouping factors.
- Predefine two primary contrasts: matched vs static and matched vs mismatched.
- Treat task success as a guardrail/non-inferiority-style outcome if the main benefit is efficiency; choose an acceptable margin only with justification.
- Report adaptation-related disruption and user control alongside completion time.

Study 3

- Evaluate estimator performance using participant-grouped cross-validation or a held-out participant set.
- Compare static and hybrid adaptive conditions on navigation outcomes and experience measures.
- Explore whether benefits vary with estimator confidence, number of switches, or user reversion, clearly labelling exploratory analyses.
- Do not claim the estimator detects true internal emotion; state that it predicts self-reported experimental context under the tested conditions.

Power and sample size

The planning targets in this document are starting points only. Use pilot data to estimate within-person variance, task-level variability, expected effect sizes, and attrition/missingness. Run a power analysis for the primary interaction in Study 1 and the matched-vs-static contrast in Study 2. If the required sample is impractical, reduce the number of layouts or tasks only with a clear rationale and preserve the central research question.

12. Validity, risks, and mitigations

Risk / threat	Why it matters	Mitigation
Learning and order effects	Participants get faster as they learn the catalogue or menu patterns.	Counterbalance layouts/tasks; use matched targets; model trial order.
Induction carry-over	A previous emotional context may influence the next trial.	Use mild inductions, neutral resets, breaks, and state re-checks.
Layout confounds	One layout may reveal more links or use different labels, not just a different structure.	Keep content inventory, labels, hierarchy, and visual style comparable.
Fatigue	Sixteen cells plus repeated tasks can make the session too long.	Pilot timing; use blocks or a balanced incomplete-block design if necessary.
Emotion misclassification	Interaction behaviours are ambiguous and person-dependent.	Use brief self-report anchors, multiple features, confidence, and held-out-participant validation.
Switching cost	Structural changes may disrupt spatial memory and learned behaviour.	Switch at task boundaries; explain changes; allow keep/revert; log disorientation.
Post-hoc mapping	Choosing the mapping after inspecting results can overfit the data.	Predefine a selection rule; freeze mapping before Study 2; report uncertainty.
One structure wins globally	There may be no useful state-specific mapping.	Treat this as a valid finding; compare global best vs state-specific policy.

Ethics and privacy

- Obtain the required institutional review/approval before collecting participant data.
- Use informed consent explaining task logging, brief emotion questions, and the right to withdraw.
- Use pseudonymous participant IDs and store the ID key separately, if one is needed at all.
- Collect only data needed for the research; set access controls and a retention/deletion schedule.
- Avoid high-intensity emotional induction and provide a debrief and contact details for study questions.
- Do not describe the system as diagnosing emotion or making psychological assessments.

13. Expected contribution and interpretation

The intended contribution is evidence about whether, when, and how navigation structure should vary with emotional context in e-commerce. The project can contribute a controlled comparison of multiple menu architectures, an empirically derived emotion-to-layout performance matrix, a causal test of structure matching, and an end-to-end prototype that accounts for imperfect state estimation and switching costs.

Possible result patterns

Observed pattern	Interpretation
Different layouts lead in different emotional contexts, and matched adaptation helps.	Supports state-contingent structural adaptation under the tested conditions.
One layout is best across all contexts.	A stable global layout may be preferable; emotion-based switching may be unnecessary.
Study 1 shows differences, but Study 2 finds no adaptive benefit.	Switching costs, task variation, or the mapping may offset the layout-specific gains.
Matched adaptation helps with known labels but not with the hybrid estimator.	Estimator reliability or transition policy is likely the bottleneck.
Users frequently revert or report disorientation.	User control and structural stability may outweigh performance benefits; redesign the transition policy.

Do not frame a null or mixed result as failure. The central scientific question is whether emotion-contingent structural adaptation is beneficial compared with a stable interface, not whether adaptation can be made to look successful.

14. Timeline and deliverables

Phase	Work	Deliverable / gate
1. Scope & design	Confirm taxonomy, four layouts, task taxonomy, state ratings, and ethics submission.	Protocol and task matrix.
2. Prototype	Build catalogue, layout renderer, task runner, event logger, and check-in UI.	Working prototype with validated logs.
3. Pilot	Run 6–8 participants; revise induction, task timing, and layout consistency.	Pilot memo and power-analysis inputs.
4. Study 1	Collect layout × emotional-context observations and analyse the mapping.	Performance matrix, uncertainty estimates, frozen mapping.
5. Study 2	Run static/matched/mismatched comparison with the frozen mapping.	Causal comparison and usability/disruption results.
6. Study 3	Add estimator and safe transitions; compare static vs hybrid adaptation.	End-to-end evaluation and estimator report.
7. Reporting	Document methods, exclusions, results, limitations, and reproducibility materials.	Final report/thesis and presentation.

15. Preregistration checklist

- Primary research question, hypotheses, and primary outcome(s) for each study.
- Exact definition and implementation of all four menu structures.
- Emotional inductions, rating items, timing, and manipulation-check rule.
- Randomisation/counterbalancing method and task equivalence plan.
- Inclusion/exclusion rules, missing data handling, and trial timeout policy.
- Study 1 mapping-selection rule and tie/uncertainty handling.
- Study 2 mismatch rule, static baseline, and primary contrasts.
- Study 3 estimator features, validation split, confidence threshold, transition policy, and fallback behaviour.
- Power analysis and justified sample sizes.

- Ethics approval, consent wording, privacy, data retention, and debriefing.

16. Literature grounding and limits

The three supplied papers provide complementary motivation, but they do not by themselves establish that any particular menu structure is best for a particular emotional state. The proposed emotion-to-layout mapping is the project's empirical question.

Eye-tracking / e-commerce menu-format paper supplied by the user. Use it to motivate systematic comparison of menu formats and objective navigation/visual-attention measures. Verify the final publication metadata and full bibliographic details before including it in a thesis bibliography. Eye tracking is not required in this protocol because the selected measurement approach is interaction signals plus brief self-reports.

Sahraoui, Vanderdonckt, and Kieffer, "Fractal Adaptive Menus: Concept, Engineering, and Use Case," EICS Companion '25. Use this work to inform the conceptual and implementation definition of the fractal/hierarchical candidate structure. The implementation in this project must still be specified precisely enough to reproduce.

Gaspar-Figueiredo et al., "Integrating Human Feedback into a Reinforcement Learning-Based Framework for Adaptive User Interfaces," EASE 2025 / arXiv:2504.20782. Use it to inform the role of human feedback and evaluation of adaptive interfaces. It does not remove the need to validate the emotion estimator or test adaptation against a static baseline.

Related external work on vertical vs dynamic menus and adaptable/adaptive menus can support background framing, but cite only after verifying the exact bibliographic records and the specific claims used. Keep source-derived claims separate from this protocol's hypotheses and design choices.

Final positioning

The revised plan now follows the original concept directly: test several menu structures across emotional contexts → determine the best-supported structure for each context → test whether switching structures based on that mapping helps → evaluate a live system using interaction signals and self-reports. Functional changes such as link prioritisation or extra guidance can remain a secondary layer, but they no longer replace structural menu adaptation as the central research contribution.